

Organic chemistry^{6th} (CHEM-317)

} 2023 {

(2)

Ans Beer's law describes the relationship between the absorbance of a solution and the concentration.

It states:

"The amount of light absorbed is directly proportional to the concentration of the solution."

$$A \propto C$$

$$A = \epsilon \cdot C$$

where ϵ is molar absorptivity constant.

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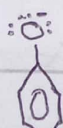
(3)

Ans Bathochromic shift: (Red shift)

A shift of λ_{max} towards longer wavelength or lower energy.
e.g. the deprotonation of phenol results in the shift of λ_{max} towards longer wavelength because the resulting ion has additional lone pairs.



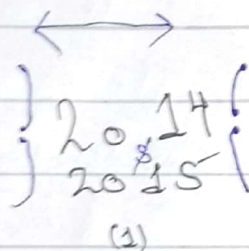
270nm



287nm

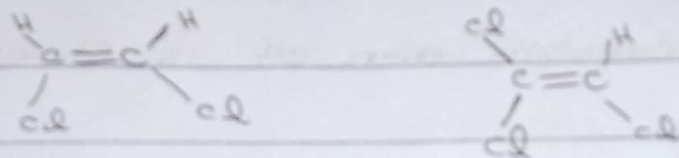
Hypsochromic shift: (Blue shift)

A shift of an λ_{max} towards shorter wavelength or higher energy.
e.g. protonation of aniline results in the blue shift because it decreases the availability of lone pair.



Ans An IR active bond is a bond that changes dipole during vibration while an IR inactive bond is a symmetric bond that doesn't change dipole during vibration. We can say that by considering the above information, cis 1,2-dichloro ethene is infrared active while trans 1,2 dichloro ethene is infrared inactive because there will be not net change in bond dipole moment. In trans product

the charges is cancelled out each other. On the other hand, cis product has a permanent dipole moment.



(2)

Ans The region between 400cm^{-1} and 1500cm^{-1} in an IR spectrum is known as the fingerprint region.
 → It usually contains a large number of peaks making it difficult to identify individual peaks.

→ In this region, small difference in the structure and constitution of a molecule results in significant changes in the absorption bands.

• ~~1500-1250 cm⁻¹~~

→ This region is valuable for identifying and characterizing organic molecules because it contains a wealth of information.

tion that is distinct for different compounds.

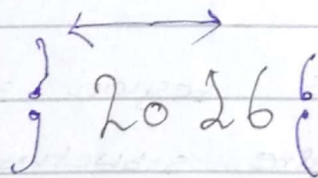


(7) When 1mol of CCl_4 is added to 2mol $\text{C}_2\text{H}_5\text{OH}$ then OH is changed. Why?

Ans Firstly CCl_4 and $\text{C}_2\text{H}_5\text{OH}$ are both are solvent. Secondly in $\text{C}_2\text{H}_5\text{OH}$ Hydrogen bonding takes place.

Hydrogen bonds are weak interaction.

So, OH group will be changed and replaced by Cl atom.



(1)

Ans Stretching vibration:

→ In this vibration, two or more bonds vibrates in and out together.

→ Stretching vibration in which bond length changes that requires more energy

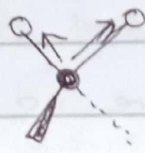
→ Distance between two atoms increases or decreases. But the atoms are in same axis.

→ stretching vibrations occur at higher energy because it involves changes in bond length.

→ There are two types of stretching vibrations:

1. Symmetric stretching:

In this type, ~~both~~ bond ~~length~~ of each atom with respect to central atom decreases or increases.



2. Asymmetric stretching:

In this type, bond ~~length~~ of one atom increases while the bond angle of other decreases.



Bending vibration:

→ It changes the angles between two bonds.

→ Bending vibration requires less energy.

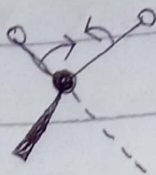
→ The position of atoms changes relative to original bond.

→ There are four types of bending vibrations.

1. Scissoring:

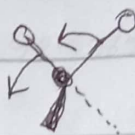
Two atoms approach each other

→ Bending vibration occurs at lower energy because it involves changes in bond angle.



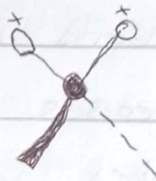
2: Rocking:

The movement of atoms takes place in same direction.



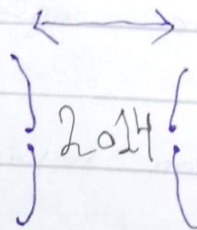
3: Wagging:

Two atoms move up and below the plane with respect to central atom.



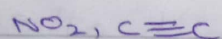
4: Twisting:

One atom moves up and other moves down the plane with respect to central atom.



Ans The term chromophore is used to denote a functional group of some other structural feature of which gives a color to compound.

For example:



There are two types of ~~chromophore~~
chromophore:

1. Independent:

If one chromophore is required to impart color. e.g. $-\text{NO}_2$, $-\text{N}=\text{N}-$

2. Dependent:

If more than one chromophore is required to impart color. e.g. Acetone

Auxochrome:

A saturated or unsaturated group with non-bonding electrons when attached to chromophore altering both wavelength as well as intensity of absorption. e.g.

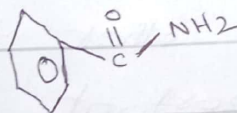
$-\text{OH}$, $-\text{NH}_2$, $-\text{I}$

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(I)

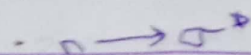
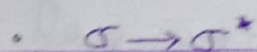
Ans



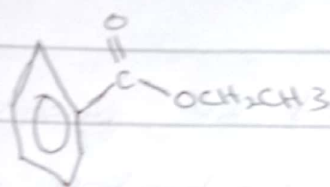
In benzamide, several types of electric transitions are possible due to excitation of electrons within the molecule.

• $\pi \rightarrow \pi^*$

• $n \rightarrow \pi^*$



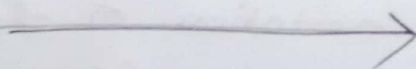
First two transition are the most relevant in benzamide and are responsible for its absorption of UV-Vis light. Other transition are less common and may not contribute significantly to its UV-Vis spectrum.



In ethyl benzoate, several transition is possible which is given below:



First two transition are important because it contribute significantly to its UV-Vis spectrum while other transition are less important.



(ii)

Ans The conjugation of a carbonyl group ($C=O$) with a $C=C$ lowers its stretching frequency due to resonance effects and increased bond strength.

* Resonance stabilization

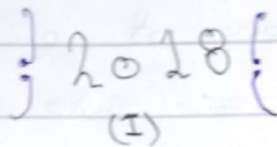
Conjugation allows for the delocalization of π electrons from the $C=C$ into the oxygen atom of the carbonyl group. This resonance results in a more even distribution of electron density in the vicinity of the $C=O$ bond. The partial double bond character is introduced, making the bond stronger and thus, lowering its stretching frequency.

* Increased bond strength

The partial double-bond character induced by conjugation increases the bond strength of the $C=O$ bond. A stronger bond requires more energy to stretch or vibrate, leading to a lower stretching frequency.

in the vibrational spectrum.

In summary, conjugation leads to resonance stabilization and increased bond strength, resulting in a lower stretching frequency of the $C=O$ bond as observed in IR spectroscopy.



Ans When a monochromatic light of initial intensity I passes through a solution in a transparent vessel some of the light is absorbed so that the intensity of the transmitted light is less than I .

The relationship between I and I' depends on the path length of the absorbing medium l and concentration of absorbing solution c . These factors are related in the laws of Lambert and Beer.

$$A = \epsilon c l$$

A = Absorbance

ϵ = Molar absorption coefficient

C = molar concentration

l = Optical path length

where as,

$$A = \log I_0/I$$

So,

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$$A = \log I_0/I = \epsilon cl$$



(11)

Ans The effect of ring size on the IR absorption in cyclic ketones can be explained in terms of hybridization of the carbons atoms involved in the carbonyl group. In ketones, the carbon atom of the carbonyl group is sp^2 hybridized. The degree of hybridization and the bond angles influence the vibrational frequencies observed in IR spectra.

In smaller cyclic ketones, the ring strain and constrained geometry result in greater p-character in the sp^2 hybrid orbitals of the carbonyl carbon.

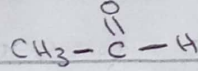
This increased p-character leads to stronger ~~and shorter~~ ~~and~~ ~~the~~ C=O bonds, resulting in higher vibrational frequencies, typically observed at higher wavenumbers around 1750 cm^{-1} . While in larger cyclic ketones, where the ring is less strained, the sp^2 hybridization of the carbonyl carbon is closer to ideal trigonal planar geometry. This results in a slightly lower p-character, longer C=O bonds, and lower vibrational frequencies, typically observed at lower wavenumbers around 1715 cm^{-1} .

So, the hybridization of the carbonyl carbon is influenced by the ring size, affecting the bond strength and bond angles. This subsequently affects the IR absorption frequencies of ketones.



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(i)

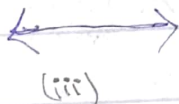


Ans

In acetaldehyde, several transition are occur. These are following:

- $\pi \rightarrow \pi^*$
- $n \rightarrow \pi^*$
- $\sigma \rightarrow \sigma^*$
- $n \rightarrow \sigma^*$

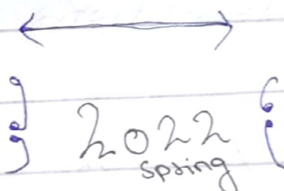
First two transitions are most prominent transitions because these are detectable easily in the UV/Vis region. While other two transitions, are not prominent.



Ans The absence of $\text{C}=\text{C}$ stretching band in the IR spectrum of ethylene can be attributed to its symmetrical molecular structure. In ethylene, the $\text{C}=\text{C}$ does not lead to a significant change in the molecule's dipole moment during stretching vibrations, resulting in a weak or absent absorption band in

the IR spectrum.

In contrast, propylene contains a $C=C$ and an asymmetrical methyl group. The presence of this asymmetrical group leads to a change in the molecule's dipole moment during $C=C$ stretching vibrations, making it IR active. As a result, propylene exhibits a distinct $C=C$ stretching band in its IR spectrum.



Ans In UV/Vis spectroscopy, the instrumentation and sample handling play a crucial role in obtaining accurate and reliable results.

* Instrumentation:

From slides

* Sample handling:

1. Sample Preparation:

The sample should be prepared properly, ensuring it is in a transparent and clean state.

Solid samples might need to be dissolved or ground into a homogeneous solution. Liquid samples should be filtered to remove particulates that could scatter light.

2. Cuvettes:

Samples are typically placed in transparent cuvette made of quartz, which allows the passage of UV/vis light.

3. Blank measurement:

A reference solution, known as the blank, is used to account for any absorption by the solvent or other components. It is measured using the same cuvette and conditions as the sample.

4. Sample Presentation:

Care must be taken to ensure that the cuvette is free from finger-prints, smudges or air bubbles, as these can effect the quality of measurements.



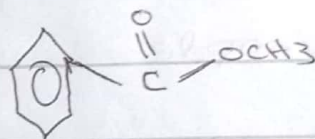
5 Scanning and Data Collection:

The spectrophotometer is set to the desired wavelength range, and the sample is measured. The instrument records the absorption or transmittance data.

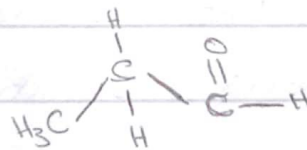
Both the instrumentation and sample handling procedures are critical to ensuring the accuracy and reproducibility of UV/Vis spectroscopic measurements.

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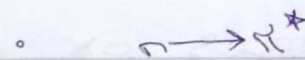
Ans



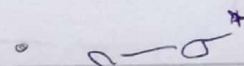
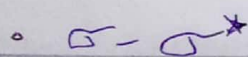
(i)



In methyl benzoate and propanal, following transitions are prominent



while following two transitions are less common.



(v)

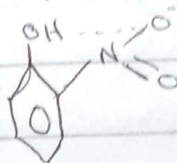
Ans Intermolecular and intramolecular hydrogen bonding significantly affect the position of absorption frequencies in a compound, particularly in IR spectroscopy.

1 Intramolecular H-bonding:

occurs when H-bonds form within the same molecule. This can lead to a shift in absorption frequencies to higher values (blue shift) compared to compounds without such bonding.

* Example:

O-nitrophenol, which has an intramolecular H-bonding between the OH group and NO₂ group.



This bonding leads to a blue shift in the absorption frequency of the OH stretching.

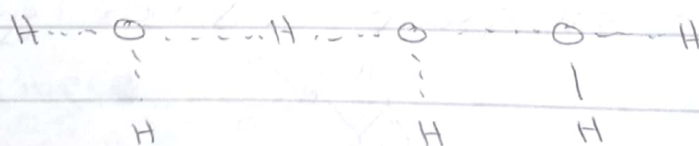
2 Intermolecular H-bonding:

occurs when molecules interact with each other through H-bonding. This can result in a

shift in absorption to lower values (red shift) compared to molecules not involved in this type of H-bonding.

* For example:

An example is water. In the gas phase, water molecules do not form extensive H-bonds with each other and O-H stretching vibration occurs at a higher frequency. However, in the liquid phase, water molecules form intermolecular hydrogen bonds, causing a red shift in O-H stretching frequency in the IR spectrum.



In summary, intramolecular H-bonding leads to blue shift while intermolecular H-bonding leads to red shift.

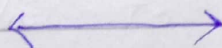


(iii)

Ans In 1-bromo-2-propene, there are several types of vibrations that can be observed in its infrared spectra. These vibrations can be categorized into stretching and bending vibrations.

There are following vibrations that occurs in this compound:

- 1: C-H stretching vibrations:
 - Alkene C-H stretching ($3000-3100\text{ cm}^{-1}$)
 - Alkyl C-H stretching ($2900-3000\text{ cm}^{-1}$)
- 2: C-C stretching vibrations:
 - Alkene C-C stretching ($1600-1680\text{ cm}^{-1}$)
- 3: C-Br stretching vibrations:
 - C-Br stretching ($500-600\text{ cm}^{-1}$)
- 4: Out of plane bending vibrations:
 - H-C=C-H bending ($900-1000\text{ cm}^{-1}$)
 - H-C-Br bending ($500-600\text{ cm}^{-1}$)
- 5: In-plane bending vibrations:
 - C-H-C bending ($1200-1300\text{ cm}^{-1}$)
 - C-H-Br bending ($600-700\text{ cm}^{-1}$)



Q How will you detect the presence of carbonyl group in aldehyde and ketones?

Ans In UV-Vis spectroscopy, the presence of a carbonyl group in aldehydes and ketones can be detected by examining their absorption spectra in the ultraviolet-visible range. Carbonyl groups have characteristic absorption bands in this region due to their $\pi \rightarrow \pi^*$ transitions.

Aldehyde typically exhibit absorption bands around 270-280 nm, while ketones have absorption bands at slightly longer wavelength, typically between 280-320 nm. By comparing the absorption peaks in the UV-Vis spectrum of a compound with these ranges, you can identify the presence of a carbonyl group and differentiate between aldehydes and ketones based on the specific wavelength of absorption.

